Lung Cancer Treatment Navigator

Final Project Document

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1. Executive Summary and High-Level overview

1.1 Executive Summary

The Problem

- **Information Overload for Oncologists:** The volume of oncology evidence is growing more rapidly than practitioners can process, leading to a significant latency gap in care.
- **Guideline Mapping Challenges:** Local and international guidelines are not easily mapped to new evidence, creating a manual burden for clinical teams.
- **The Bottieneck: Triage & Contextualization:** The core delay in evidence-based medicine occurs during the manual sorting and situational analysis of new clinical data.

Our Solution

- **Deterministic AI Assistant:** A rule-based system designed for predictable, reiiable output rather than generative uncertainty.
- **Evidence Retrieval from PubMed:** The system automatically retrieves and ranks relevant evidence specifically for Non-Small Cell Lung Cancer (NSCLC).
- **Mapping to ESMO Guidelines:** Evidence is systematically compared against the European Society for Medical Oncology standards for clinical alignment.
- **Automated Labeling:** Results are flagged as Aligned (Silent) when they match guidelines or Conflict (Aiert) when a discrepancy is detected.

Why AI?

- **Semantic Evidence Retrieval:** AI enables the understanding of context and meaning within medical literature beyond simple keyword matching.
- **High Volume & Update Rate:** Machine learning systems can handle the massive throughput and constant updates of oncology research that exceed human capacity.
- **Pattern-Based Guideline Mapping:** AI identifies structural patterns within clinical guidelines to automate the comparison process with new evidence.

Key Innovations

- **Deterministic, Rule-Based System:** Logic follows a strict set of rules to ensure consistency and explainability in medical decision support.
- **Embedded Evaluation Framework:** The system includes a built-in mechanism for continuous assessment of its own accuracy and relevance.
- **Human-in-the-loop Oversight:** Every output is subject to a mandatory safety check by a qualified oncologist to maintain clinical accountability.

What We Don't Do

- **No Diagnosis or Treatment Selection:** The tool is an evidence-triage assistant, not a primary diagnostic or prescriptive medical device.
- **No Free-Text Generation:** The system avoids the risks of "hallucinations" by not using generative free-text capabilities.
- **No Patient Data:** The AI operates on clinical evidence and guidelines without accessing or processing individual patient-identifiable information.

1. The Problem

- Global oncology research grows faster than clinicians can track.
- Causes delays in guideline integration.

The Evidence Latency Gap
This causes outdated treatment protocols.

2. Our Solution

- Retrieves PubMed evidence.
- Maps to ESMO guidelines for NSCLC.
- Uses transparent rules for ranking.

Evidence Ranking Score (ERS)

Evidence Ranking Score (ERS)

Evidence Strength (e.g., RCTs vs. Case Series)	0–20 Points
Dataset Robustness (Sample Size N)	0–15 Points
Source Credibility & Recency	0–25 Points

3. Why AI?

- Uses transparent rules for ranking to ensure auditability and safety.

Determinism: Auditability & Safety
Uses transparent rules for ranking.

Generative: Black Box
Not used to ensure auditability in high-risk clinical settings.

4. Key Innovations

Two-Layer Evaluation Framework
- Separates technical system functionality from clinical correctness.
- Ensures measurable and interpretable performance.

Smart Evidence Labeling
- Labels research as 'Aligned,' 'Guideline-Silent.' or 'In Conflict' based on ESMO stance.

Strict Safety Boundaries
- Ensures system output remains within established clinical guardrails.

5. What We Don't Do

- Does not provide diagnosis, prescribing, or autonomous medical advice.

No Diagnosis/ Prescribing

No Free-form Generative Advice

No Autonomous Decisions

1.2 Project Overview

This project proposes a deterministic AI-driven evidence triage and guideline mapping assistant for Non-Small Cell Lung Cancer (NSCLC).
The system addresses the growing evidence latency gap in oncology: the time delay between publication of new high-quality research and its structured integration into clinical decision-making contexts.

The assistant:

- Retrieves relevant NSCLC evidence from PubMed (evidence library)
- Ranks it using a deterministic Evidence Ranking Score (ERS)
- Maps evidence clusters to ESMO guideline topics
- Labels evidence as:
    - Aligned
    - Guideline-Silent
    - In Conflict

The system does not provide treatment recommendations. It provides structured, auditable evidence navigation.

1.3 Main Findings

Through our lifecycle analysis we conclude:

- Deterministic architectures are more appropriate than generative advisors in high-risk clinical domains.
- Evaluation design must be treated as a first-class architectural component.
- Structured inputs significantly reduce ambiguity and model risk.
- Scope discipline dramatically improves feasibility and defensibility.
- Responsible AI principles can be embedded directly into system design

2. Problem and proposed solution

2.1 The problem

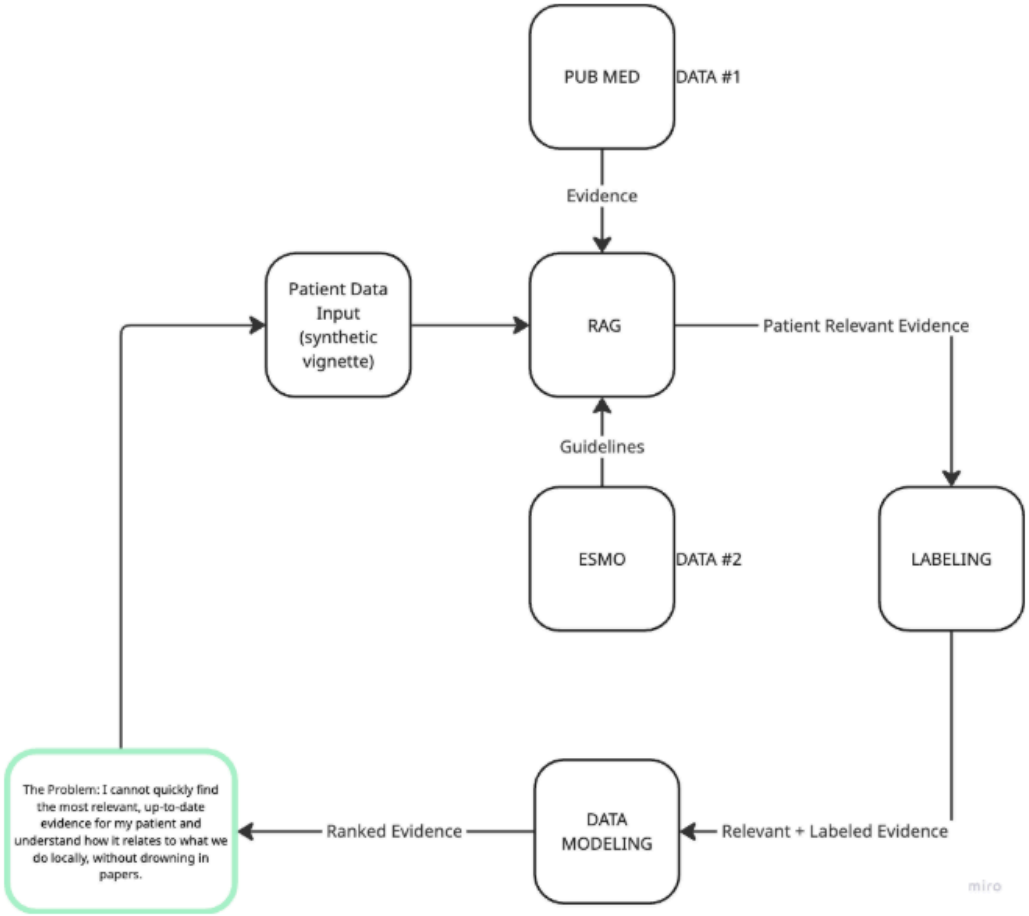
Oncologists make high-stakes decisions while global oncology evidence grows faster than any individual can track. The practical pain is not “lack of evidence,” it is **overload plus uncertainty about what matters for this specific patient, right now, in my local context**.

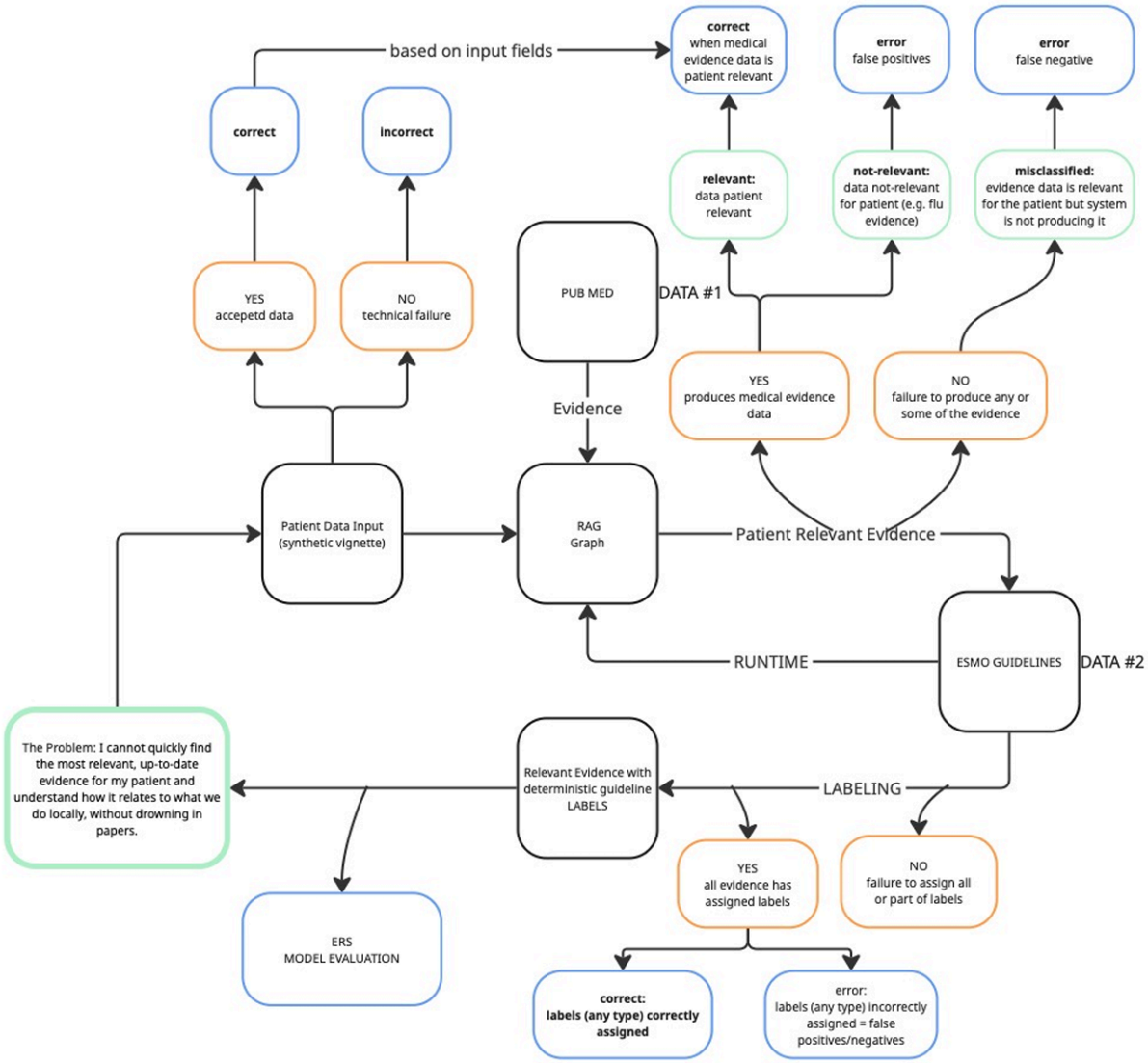
2.2. The goal

A deterministic decision-support assistant that **triages recent global evidence** for a given lung cancer patient scenario and **maps it to ESMO guideline context**, making it fast to see:

- what evidence is most relevant,
- what guideline topic it relates to,
- what assumptions and missing context limit interpretability.

2.3 Our MVP





2.4 User journey (deterministic)

1. Oncologist enters a structured patient snapshot + clinical question (treatment, tests, follow-up).
2. The tool retrieves the most relevant systematic-review evidence (and linked studies when available) from PubMed.
3. The tool ranks and clusters evidence deterministically using transparent rules (recency, study type, population match, intervention match, outcomes).
4. Tool maps each evidence cluster to ESMO guideline topics and labels it:
    - **Aligned** (supports direction already reflected in guideline)
    - **Guideline-silent** (not covered or not yet updated)
    - **Conflicts** (suggests direction that contradicts guideline)
5. The tool outputs the ranked evidence list, mapping labels, and the guideline reference points used.

2.5 Scope limitations

2.5.1 In scope (MVP)

- **Country context:** International (guideline mapping uses European ESMO guideline sources). We focus on ESMO due to affordable data structure and availability, postponing any local feasibility to layer 1.
- **Disease:** Lung cancer (we start with NSCLC only)
- **Core capability:** Evidence triage + mapping (no “best treatment” claims)

2.5.2 Out of scope (explicit non-goals)

- Diagnosis support
- Autonomous prescribing or “best treatment” selection
- Reimbursement coverage and hospital formulary availability
- All cancers, all countries
- Free-form generative medical advice (outputs are structured and traceable)

3. Data Strategy

For the MVP, we use two data sources. One for evidence retrieval (PubMed) which is then mapped against existing guidelines (ESMO).

3.1 Evidence Data (PubMed)

PubMed (https://pubmed.ncbi.nlm.nih.gov/) is a free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health globally and personally. We use PubMed for systematic-review evidence retrieval and updating, via their methods and API.

Data retrieved:

- Study metadata
- Abstract
- Publication date
- Journal source
- Study type tags

Filtered to:

- Systematic Reviews
- Randomized Controlled Trials

3.2 Guideline Data (ESMO)

ESMO (https://www.esmo.org/) is the European and International guideline database as the home of multidisciplinary oncology guidelines. ESMO complies with EU regulations and is a reliable guideline database. We create a structured topic catalog derived from these guidelines.

Example topics:

- First-line metastatic NSCLC
- Second-line EGFR-mutated NSCLC
- Adjuvant therapy settings

Each topic includes:

- Applicability rules
- Intervention tags
- Guideline stance

3.3 Evaluation Data

We use 15 frozen synthetic patient vignettes, which are short, synthetic patient scenarios used for testing. It includes the minimum inputs required for evidence triage and mapping. A sample vignette is:

- Patient: adult, Ireland
- Diagnosis: NSCLC
- Stage: locally advanced
- Biomarkers: EGFR unknown, ALK negative, PD-L1 60%
- Performance status: 1
- Clinical question: “What evidence is most relevant for first treatment choice and what guideline topic does it map to?”

Thus, vignettes include structured features, known, relevant evidence and mapping labels. This serves as a regression test set.

4. Data Conditioning

4.1 Structured Inputs

- Lung cancer type (fixed NSCLC for MVP)
- Stage, line of therapy (captured, not used to exclude cases)
- Key biomarkers (yes/no/unknown)
- Performance status
- Clinical question type: fixed for MVP: **treatment evidence**. Post MVP: **tests**, **follow-up**

4.2 Clinical Relevance Gate (hard stop)

An evidence item is eligible for ranking only if it matches the vignette on:

1. Disease match: evidence.population_tags.disease includes NSCLC
2. Setting match: evidence.population_tags.setting matches vignette.disease_setting OR is “mixed” and still applicable

3. Histology match: evidence.population_tags.histology is vignette.histology OR “all_nsclc” OR “mixed”
4. Biomarker compatibility:
    - if evidence is biomarker-specific (example: EGFR-mutant only), then vignette must satisfy that biomarker condition
    - if evidence biomarker applicability is “unspecified”, it passes gate but receives no extra match benefit later

If an item fails the gate, it is not ranked. It may be shown under “Secondary references (not applicable)” with the reason.

4.3 Guideline Compliance Gate

For the second gate, an internal “topic catalog” aligned to vignette fields (topic applicability rules and intervention tags) is created. Guideline stances are recorded where available(match / does_not_match / no_matching_data) to support aligned vs conflict vs silent labels.

We are still considering:

- Exact extraction and tagging method for PubMed (pure keyword rules vs lightweight NLP)
- How much manual labeling is needed to establish a reliable Guideline topic catalog and reference mapping for evaluation.

5. Data Modeling

We employ a deterministic pipeline to ensure safety and auditability. LLMs are used only for bounded tasks (e.g. query expansion), never for medical reasoning.

Rank NSCLC evidence items deterministically for a given patient vignette and map each ranked item to the **Guideline Topic Catalog**, producing:

1. a ranked “Top Evidence” list (triage), and
2. a mapping label per item: **aligned / guideline-silent / conflict** (mapping).

This MVP does not recommend treatment. It ranks and labels evidence relative to guideline topics.

5.1 Evidence Ranking Score (ERS) Algorithm

Evidence is ranked on a 0-60 scale based on the following weighted factors:

Factor	Scoring Criteria	Points
Evidence Strength & Methodology	Guideline (20), Systematic Review (18), Ph3_RCT (16), Ph2_RCT (13), Prospective_obs (10), Retrospective (6), Case_series / Expert_opinion (2)	0–20
Database Robustness	N≥300 (15), 100-299 (12), 50-99 (9), 20-49 (6), <20 (3). If relevant_N is null: *Default: 6 (mid, conservative)*	0–15
Source Credibility	Guideline Body (15), High Impact (12), Specialty (9), Preprint (5), Industry (2). if source_category is null: default 6 Default: 6	0–15
Recency of publication	≤3 yrs (10), 4–6 yrs (6), 7–10 yrs (3), >10 yrs (0)	0–10

Sorting: descending ERS-MVP, with deterministic tie-breakers:

1. higher A wins
2. newer year wins
3. larger N wins
4. otherwise keep both and cluster by topic

Default inclusion threshold:

- include in “Top Evidence” if ERS-MVP ≥ 30
- show below threshold as “Secondary reference”

A) Evidence strength & methodology (0–20)

- guideline: 20
- systematic_review: 18
- phase3_rct: 16
- phase2_rct: 13
- prospective_obs: 10
- retrospective: 6
- case_series / expert_opinion: 2

B) Dataset robustness (0–15) using relevant_N

- N ≥ 300: 15
- 100–299: 12
- 50–99: 9
- 20–49: 6
- < 20: 3
- if relevant_N is null: default 6 (mid, conservative)

C) Source credibility (0–15)

- guideline_body: 15
- high_impact_journal: 12
- specialty_journal: 9
- preprint: 5
- industry_whitepaper: 2
- if source_category is null: default 6

D) Recency (0–10) using publication_year (relative to current year)

- ≤3 years: 10
- 4–6 years: 6
- 7–10 years: 3
- 10 years: 0

ERS-MVP = A + B + C + D

Sorting: descending ERS-MVP, with deterministic tie-breakers:

5. higher A wins
6. newer year wins
7. larger N wins
8. otherwise keep both and cluster by topic

Default inclusion threshold:

- include in “Top Evidence” if ERS-MVP ≥ 30
- show below threshold as “Secondary reference”

5.2 Topic Mapping Logic

Using the chosen topic’s guideline_stance:

- **Aligned:** Topic stance is recommended or conditional, and evidence supports that intervention category in an applicable population
- **Conflict:** Topic stance is do_not_recommend, but evidence suggests benefit for that same intervention in applicable population
- **Guideline-silent:** Topic stance is not_covered, OR no topic match was found

For the MVP, “supports” is implemented deterministically as:

- evidence_type guideline/systematic_review/phase3_rct is considered “supporting” if it matches topic intervention_tags and passes gate
- you can refine later with directionality extraction, but MVP stays conservative

5.3 Outputs

For each ranked evidence cluster:

- Evidence items (systematic reviews first) and linked underlying studies when available
- Deterministic relevance score factors (why it is ranked here)
- Guideline mapping label: aligned / guideline-silent / conflicts
- Guideline reference pointer(s) used to make that label
- Key assumptions and limitations (example: missing biomarker prevents subtype-specific interpretation)

6. Evaluation

The evaluation framework follows a **two-layer architecture**:

1. System Integrity (Technical Layer) -> "*Did the system function?"*
2. Clinical & Logical Correctness (Evaluation Layer) -> "*Was the output correct?"*

This design ensures:
- Operational failures are separated from clinical misclassifications.
- Deterministic model logic is not confused with clinical correctness.
- Errors are measurable, interpretable, and attributable to specific stages.

6.1 Layer 1 - System Integrity (Technical Layer)

This layer verifies whether the system operates structurally as intended. These checks assess system integrity, not medical accuracy.

Key Principle: A "Yes" means the system functioned. It does not imply correctness.

Step 1: Input Acceptance

Question: Did the system successfully parse structured vignette data?
- Yes = Required fields parsed and stored
- No = Input rejected or improperly processed

Step 2: Evidence Retrieval

Question: Did the system generate medical evidence output? (evaluates output existence, not relevance.
- Yes = Evidence items are generated
- No = No evidence retrieved

Step 3: Label Assignment

Question: Did the system assign exactly one valid label?*
- Yes = One valid label assigned

- No = Missing, multiple, or invalid label

**Allowed labels*
- *Aligned (supports direction already reflected in guideline)*
- *Guideline-silent (not covered or not yet updated),*
- *Conflict (suggests direction that contradicts guideline)*

6.2 Layer 2 - Evaluation Layer (clinical layer)

This layer evaluates outputs against **predefined reference standards** frozen prior to testing. Evaluation Layer begins only after all Technical checks return “Yes.”

Evaluation judgments use: Correct and Error

Step 1: Evidence Relevance Assessment

Question: Is the produced evidence relevant to the patient vignette? (NSCLC scope, stage, ECOG slice, biomarkers, treatment question)?
- Correct = system matches reference relevance
- Error = misclassification (False Positive or False Negative)

Relevance reference:
- Frozen vignette definition
- Expert-reviewed relevance annotations

Definitions:
- FP: System marks Relevant but reference says Not Relevant
- FN: System marks Not Relevant but reference says Relevant
- Silent (Abstention): Tracked separately, not automatically counted as FN

Risk principle: FN prioritized due to omission risk.

Step 2: Guideline Mapping Accuracy

Question: Does the assigned label (Aligned / Silent / Conflict) match the predefined guideline mapping table (ESMO-based)?

Multi-class (3×3) evaluation preserves semantic distinction between:
- Aligned (supported)
- Silent (not covered)
- Conflict (contradicted)

Human-in-the-loop validation is required at this stage for Phase 2.

Step 3: Citation Validity

Question: Does the citation provided support the claim made?
- Correct = Citation directly supports output

- Error = Citation missing, irrelevant, or unsupported

Step 4: Model Logic Validation (Deterministic Rule Audit)

This step audits implementation fidelity, not clinical judgment. Target: **100% fidelity**

Step 4A. Clinical Relevance Gate

Question: Were eligibility rules applied exactly as specified? (e.g. biomarker, histology, disease filters)

- Correct = Gate behavior matches model specification
- Error = Incorrect inclusion or exclusion

Step 4B. Evidence Ranking Score (ERS) Audit

ERS = Evidence Ranking Score (0–60)

Question: Was ERS computed exactly according to the predefined A/B/C/D scoring rules?

- Correct = Score and ranking match deterministic specification
- Error = Miscalculated component, incorrect total, threshold misapplication, or tie-break violation

Audit checks:

- Correct assignment of **Evidence Strength** points (A) - > 0-20
- Correct Dataset **Robustness** scoring (B) -> 0-15
- Correct Source **Credibility** scoring (C) -> 0-10
- Correct **Recency** scoring (D) -> 0-15
- Correct handling of null/default values
- Correct total score computation
- Correct application of threshold (≥30 for “Top Evidence”)
- Correct deterministic tie-break logic

Step 4C. Ranking Consistency & Invariance Validation

Purpose:Verify if monotonic scoring rules are respected, for example:

1. If Evidence A has a higher A-score (Evidence Strength) than Evidence B and all other components equal, ERS_A ≥ ERS_B.
2. Increasing sample size (N) must not decrease ERS.
3. More recent publication year must not decrease ERS.
4. Higher total ERS must always rank above lower ERS (no sorting inconsistencies).
5. Threshold rule (≥30) must always classify evidence correctly into Top vs Secondary.

Evaluation:

- Correct = All ranking invariants satisfied.
- Error = Any invariant violated + description of error.

Step 4D. Label Logic Consistency

Question: Was aligned/conflict/silent assigned strictly per specification?

- Correct = Label matches deterministic mapping logic
- Error = Logical misapplication of stance rules

6.3 Error Analysis Framework

Errors are calculated **only in the Evaluation Layer.**

The Evaluation Layer compares system outputs against a **predefined reference standard**, which is frozen prior to testing and varies by evaluation step

Step	What Is Being Judged	Required Reference
Vignette interpretation	Input mapping correctness	Frozen vignette definition
Evidence relevance	Applicability to patient	Expert-reviewed relevance annotation
Guideline alignment	Label correctness	Predefined guideline mapping table
Citation validity	Traceability	Human verification of citation support

6.3.1 Evidence Relevance Classification

Reference Standard
Relevance annotations are defined using:

- The frozen vignette definition
- Expert-reviewed relevance mapping (Team B + clinical validation)

For relevance classification:

- False Positive (FP): System marks Relevant but reference says Not Relevant
- False Negative (FN): System marks Not Relevant or Silent but reference says Relevant
- Abstention (Silent Case): System marks *Silent* but reference says *Relevant*.
  Abstentions are tracked separately from FN.
  They are not automatically counted as False Negatives but may be included in a combined “Miss Rate” if reported.

6.3.2 Guideline Alignment Classification

Reference Standard
Guideline alignment is determined using a predefined mapping table based on the selected guideline database (ESMO), validated prior to evaluation.

For guideline alignment classification:

- False Positive (FP): System marks Aligned but reference says Conflict or Guideline-Silent
- False Positive (FN): System marks Conflict or Guideline-Silent but reference says Aligned

Domain risk asymmetry:
In clinical decision-support systems False Negatives (missed relevant evidence) carry higher clinical risk than False Positives. Therefore, minimizing FN is prioritized over maximizing precision. The primary optimization objective is to Maximize Recall for the Relevant class.

- Recall (Relevant class) is prioritized
- Target FN rate ≤ 5% (MVP validation goal)
- False Positive Rate ≤ 10–15%

6.4 Performance Metrics (classification-based)

Performance evaluation is divided into two domains (must not be combined)

1. Clinical Classification Performance
2. Deterministic Model Integrity Performance

6.4.1 Clinical Classification Metrics*

Core Metric - Misclassification Rate (0-1 Loss)

Question: "How often did the system get the label wrong?
Misclassification Rate = Number of incorrect predictions/ total prediction

Classification 0-1 loss (defines what error is)

- 0 = if prediction is correct
- 1= if prediction is incorrect

**Because the system produces categorical outputs (Relevant / Not Relevant / Silent, Aligned / Silent / Conflict), evaluation uses classification metrics, not regression metrics.*

6.4.2 Domain-Specific Confusion Matrices

Confusion matrices are not aggregated across domains.

Evidence Relevance (Binary 2×2 Classification)
Metrics:

- **False Negative Rate** (FN ≤ 5%): *The system fails to mark relevant evidence as relevant.* In clinical decision support, FN is prioritized due to higher risk of missed relevant evidence.
- **False Positive Rate** (FP ≤ 10–15%): *The system marks evidence as relevant/aligned when it is not.*
- **Overall Accuracy** ≤ 10–15%
- **Recall (Recall ≥ 95%)**: *Measures ability to capture all truly relevant evidence*

- **Precision:** *Measures reliability of "Relevant" prediction*
- **F1 Score:** *Balances precision and recall*

A standard 2×2 confusion matrix is used:

	Reference Relevant	Reference Not Relevant
Predicted Relevant	True Positive (TP)	False Positive (FP)
Predicted Not Relevant	False Negative (FN)	True Negative (TN)

Guideline Mapping (3×3 Multi-Class Classification)
Metrics:

- Overall mapping error ≤ 10–15%
- Per-class error rates reported

	Reference Aligned	Reference Silent	Reference Conflict
Predicted by the system - **Aligned**	Correct	Error	Error
Predicted by the system - **Silent**	Error	Correct	Error
Predicted by the system - **Conflict**	Error	Error	Correct

Citation Validity (Binary 2×2 Classification: Valid/Invalid)
Metrics

- Citation Error Rate ≤ 10–15%
- Missing Citation Rate (FN / Required Cases)
- Misleading Citation Rate (FP / Provided Citations)

Deterministic citation logic (if rule-based): 100% fidelity expected.

Citation validity evaluates whether the system's cited sources directly support the specific claim made in the output. Where:

- True Positive (TP): Citation is present and directly supports the claim.
- False Positive (FP): Citation is present but does not support the claim (misleading or irrelevant).
- False Negative (FN): Citation is required but missing.
- True Negative (TN): Citation correctly omitted when not required.

6.4.2 Deterministic Model Integrity Metrics

(Layer 2 – Step 4)

All require 100% fidelity:

- Gate Accuracy: *% of cases where gate inclusion/exclusion matches specification.*
- ERS Calculation Accuracy: *% of cases where:*
    - *A/B/C/D components are correctly assigned*
    - *Total ERS computed correctly*
    - *Threshold (≥30) applied correctly*
    - *Tie-break logic applied correctly*
- Ranking Invariance Satisfaction: *% of ranking invariants satisfied (e.g., higher A score must not rank below lower A when other factors equal).*
- Topic Mapping Accuracy: *% of cases where topic selection matches deterministic overlap rules.*
- Label Logic Accuracy: *% of cases where aligned / conflict / silent labels match stance rules.*

6.4.3 Generalizations Controls

Applies to clinical metrics only.

- Data split: *Leave-One-Out Cross-Validation (LOOCV) or 70/30 split*
- No data leakage train/test dataset
- Probability Threshold tuning only on validation not test data (Default:Threshold = 0.5
- Ridge (L2) regularization for polynomial model. Why? Given: 6 input features, Polynomial expansion (degree 6), Small dataset there's a risk of high variance / overfitting

Regression metrics (MSE, MAE, R^2) are inappropriate because the task is classification, not regression.

6.4 Governance Statement

This evaluation framework:

- Separates system integrity, model logic fidelity, and clinical correctness
- Uses classification error aligned with generalization theory
- Explicitly measures misclassification (0–1 loss)
- Adds deterministic model integrity validation
- Incorporates bias–variance controls (data split + regularization)
- Avoids inappropriate regression metrics
- Prioritizes minimizing False Negatives due to medical risk asymmetry

6.5 Performance Metrics Overview (Multi-Matrix Architecture)

Domain	Classification Structure	Metrics Reported	MVP Target	Purpose

Evidence Relevance	**Binary (2×2 Confusion Matrix)** Relevant / Not Relevant	Misclassification Rate (0–1 loss) False Negative Rate (FN) False Positive Rate (FP) Recall Precision F1 Score Accuracy	FN ≤ 5% FP ≤ 10–15% Overall ≤ 10–15%	Control clinical omission risk (missed relevant evidence prioritized)
Guideline Mapping	**Multi-Class (3×3 Confusion Matrix)** Aligned / Silent / Conflict	Overall Mapping Error Rate Per-Class Error Rates (Aligned, Silent, Conflict) Optional Binary View (Aligned vs Not Aligned)	Overall ≤ 10–15%	Preserve semantic distinctions between alignment, silence, and conflict without collapsing meaning
Citation Validity	**Binary (2×2 Confusion Matrix)** Valid / Invalid	Citation Error Rate Missing Citation Rate (FN) Misleading Citation Rate (FP)	≤ 10–15%	Ensure traceability and justification independent of clinical mapping
Model Logic – Gate	Rule Audit (Binary)	Gate Accuracy Rate	100%	Validate deterministic eligibility logic
Model Logic – ERS Calculation	Rule Audit (Binary)	ERS Component AccuracyTotal Score AccuracyThreshold Integrity	100%	Ensure Evidence Ranking Score fidelity
Model Logic – Ranking Invariance	Rule Audit (Binary)	Invariance Satisfaction Rate	100%	Ensure monotonic and stable ranking behavior
Model Logic – Topic & Label Logic	Rule Audit (Binary)	Mapping Logic AccuracyLabel Logic Accuracy	100%	Validate deterministic stance assignment

7. Responsibility and Governance

In our MVP, safety is enforced by constraining system behavior rather than having the system make or suggest medical decisions. Safety boundaries are implemented through fixed vocabularies, structured inputs, explicit uncertainty language, and vignette-based evaluation, with human reviewers validating compliance.

7.1 Safety Boundaries - System Constraints

We define explicit safety boundaries that limit what the system can and cannot do.
Each boundary is **implemented through concrete artifacts**, not abstract principles.

Safety boundary	How it´s enforced	Where it´s visible
Clinical authority	Descriptive-only output language	Evidence summaries, relevance scoring
Guideline interpretation	Restricted label set (aligned/silent/conflict)	Mapping_Accuracy sheet
Knowledge completeness	Mandatory uncertainty language	Output footer, one-pager
Input assumptions	No inference; explicit checks	vignette metadata sheet
Scope limitation	Treatment evidence only	Scoring sheet structure
Evaluation limits	Behaviour-based metrics only	Misclassification definition
Evaluation control	Vignette-based regression	Reused scoring sheet

Each boundary is enforced via:

- Fixed output templates
- Restricted label vocabularies
- Structured input fields
- Yes/No evaluation checks in the scoring sheet

Boundary compliance is verified through human review using the scoring sheets; enforcement is therefore socio-technical, not fully automated.

7.2 Update and Non-Regression Policy - System Changes

As evidence and system logic evolve, we require that updates do not silently degrade previously validated behavior.

What Can Change (with controls)

Component	Examples
Evidence Corpus	New PubMed Entries
Parsing & Retrieval logic	Improved keyword extraction
Mapping heuristics	Better guideline alignment rules
Output formatting	Clearer summaries, better UX

What Is Frozen for MVP (No change without re-approval

Component	Reason
Scope (NSCLC, treatment evidence only)	Prevent scope creep
Input fields	Maintain vignette comparability

Label vocabulary	Ensure consistent evaluation
Safety language templates	Prevent clinical overreach

7.3 Non-regression Mechanism

Fixed Evaluation Set

- A frozen set of 10–15 NSCLC patient vignettes
- No unknown fields
- Represents diverse biomarker and disease settings

Mandatory Re-Evaluation

Every update must be re-scored on the same vignette set using the same scoring sheets:

- Vignette Metadata Alignment
- Evidence Relevance
- Mapping Accuracy
- Citation Validity
- Misclassification flags

7.4 Non-regression Criteria

Hard Stops (Update Blocked)

- Introduction of recommendation language
- New guideline or approval claims
- Increase in misclassification errors
- Removal of uncertainty disclosures

Soft Review (Allowed with Documentation)

- Changes in evidence relevance
- Shifts in mapping labels (e.g., aligned → silent)
- Addition of new evidence without loss of prior context

All soft changes must be documented. Each update produces a short update record describing what changed, why, and the observed non-regression impact.

8. Project evolution and our learning

At the start of our project, our ambition was broad. To provide oncologists with an AI-powered assistant, capable of feeding the latest research from around the globe into more effective, consistent, and informed treatment decisions and standardised cancer care quality. However, early feedback from the faculty highlighted three critical risks: enormous

data volume, challenges in evaluation, and local bias. These comments made us reassess not just our scope, but also our understanding of what makes an AI/ML initiative viable in high-stakes domains.

Our first major learning was that ambition without evaluation discipline is fragile. We recognized that a broad oncology support tool would be difficult to validate rigorously, especially if it relied on stochastic, generative outputs. In response, we narrowed the scope to Non-Small Cell Lung Cancer (NSCLC) - after the kind consultation of a colleague in our cohort - a domain that is clinically significant yet bounded enough to allow for structured modeling and testing. This pivot was not merely a reduction in scope; it was a strategic decision to prioritize credibility, feasibility, and measurable performance.

A second fundamental shift concerned system architecture. Our initial concept leaned toward a generative advisory system that could interpret evidence and provide clinical suggestions. However, we learned that in high-risk medical environments, stochastic outputs introduce unacceptable ambiguity in both evaluation and governance. The question “What is the correct output?” becomes difficult to answer when responses are probabilistic. We therefore pivoted to a deterministic, rule-based evidence triage engine. Large language models are now used only for tightly bounded tasks, such as query expansion, and never for medical reasoning. We learned that in certain domains, structured determinism is most valuable.

Another key insight emerged around the issue of local bias. Our original plan involved mapping evidence to local Dutch hospital guidelines. Through discussion and reflection, we realized that this approach would limit scalability, introduce institutional bias, and complicate evaluation, besides the fact that translation would have been another issue. We therefore transitioned to mapping evidence against structured topics derived from the European Society for Medical Oncology guidelines (ESMO). This shift allowed us to anchor the system in a standardized, internationally recognized reference framework. By doing so, we increased portability, reduced contextual variability, and created a clearer “ground truth” for evaluation.

We also deepened our understanding of data governance. Initially, the abundance of medical literature appeared to be an opportunity. Over time, we recognized that uncontrolled data volume increases system complexity and undermines reliability. We therefore imposed strict constraints: a defined PubMed-based evidence corpus, structured vignette inputs instead of free text, and a frozen set of synthetic evaluation cases for regression testing. We came to understand that data conditioning is not a technical preprocessing step; it is a risk management strategy.

Another significant learning concerned the evaluation architecture. Early in the project, our attention was drawn toward modeling techniques and feature design. As the project evolved, we realized that the defining characteristic of a credible AI system is not its modeling sophistication, but its evaluation structure. We designed a two-layer evaluation framework that separates system integrity from clinical correctness. We introduced confusion matrices, recall targets, citation validity checks, and non-regression testing against a frozen vignette set. Evaluation became a core architectural component rather than a downstream metric.

Over time, our project also evolved from feature development toward safety engineering. We made explicit what the system does not do: it does not diagnose, prescribe, or recommend treatment. It structures and maps evidence. We embedded transparency through traceable citations, visible scoring logic, and explicit mapping rules. Human oversight remains mandatory. This deliberate boundary-setting reflects an understanding that responsible AI is not an add-on but a design principle.

Finally, we reframed our project at a strategic level. What began as an “AI assisting doctors in treatment decisions” became an infrastructure tool for structured evidence triage and guideline contextualization. This reframing significantly reduces legal risk, increases institutional adoptability, and aligns with enterprise AI deployment realities.

In summary, our project journey reflects three overarching lessons. First, scope discipline is a strategic strength, not a limitation. Second, evaluation architecture defines credibility. Third, in high-stakes domains, deterministic and auditable systems may be more appropriate than highly expressive generative ones. These insights extend beyond this specific use case and shape how we now think about AI/ML initiative design more broadly.

9. The way forward: commercialisation potential and challenges ahead

The current MVP establishes a defensible architectural blueprint for deterministic evidence triage and guideline mapping in NSCLC. The next phase would focus on validation, refinement, and controlled institutional deployment.

In the short term, priorities include expanding and refining the structured guideline topic catalog, strengthening evidence tagging logic, and enlarging the evaluation vignette set. A formal clinical advisory board would be essential to review rule logic, mapping criteria, and scoring weights. This step ensures medical robustness and strengthens stakeholder trust.

A prototype user interface could be developed, e.g. for for tumor board settings and tested in observational mode, where outputs are reviewed but not operationally relied upon. This allows workflow integration testing without clinical risk.

In the medium term, expansion to adjacent oncology domains may be considered. However, each new cancer type would require independent validation, domain-specific topic catalogs, and fresh regression testing to maintain credibility.

In the long term, the system could evolve into continuously monitored evidence infrastructure, where new publications and guideline updates automatically trigger structured re-evaluation.

9.1 Commercialization Potential and value proposition

If the above further development plans are positive, there might be commercial opportunities to enhance institutional efficiency, transparency, and risk management.

For hospitals and oncology centers, the value proposition could be the following:

- Reduced time spent on manual evidence review
- Improved tumor board preparation efficiency
- Structured documentation of guideline alignment
- Transparent audit trails for compliance purposes

The solution is best positioned as enterprise clinical infrastructure rather than a standalone decision-making tool. Integration via API into existing clinical systems could further enhance adoption.

The competitive differentiation lies in deterministic architecture, transparent rule logic, and embedded evaluation safeguards.

9.2 Challenges Ahead

The following challenges remain before we could consider real-world deployment:

- **Clinical Validation:** Synthetic vignette testing must be complemented by real-world validation using anonymized patient data and expert review. Without external validation, adoption barriers will remain high.
- **Regulatory Considerations:** Depending on jurisdiction and positioning, the system may fall under Software as a Medical Device (SaMD) regulations. Careful scope definition — emphasizing evidence structuring rather than treatment recommendation — will be critical to manage regulatory burden.
- **Guideline Maintenance:** Clinical guidelines evolve regularly. Sustained credibility requires a structured update protocol to ensure mapping logic remains current. Ongoing governance will be necessary.
- **Data and Evidence Ambiguity:** Medical literature is heterogeneous, and abstract-level filtering may occasionally be insufficient. Balancing automation with reliability remains a key design tension.
- **Adoption and Trust:** Successful deployment depends not only on technical robustness but also on clinician trust. Transparent logic, clear boundaries, and strong clinical sponsorship will be essential for integration into real workflows.

9.3 Strategic Outlook

This project demonstrates that impactful AI in medicine does not require autonomous decision-making. Structured, deterministic systems that reduce cognitive overload and increase transparency can deliver meaningful institutional value while maintaining safety.

If developed with rigorous validation, careful regulatory positioning, and disciplined scope expansion, the NSCLC Evidence Triage and ESMO Mapping Assistant has the potential to evolve from an academic blueprint into a commercially viable clinical intelligence platform.

More broadly, the project reinforces a central lesson of the AI/ML Initiative Life Cycle: long-term value emerges not from algorithmic sophistication alone, but from disciplined problem framing, evaluation rigor, and responsible deployment design.